

PALMS System Design

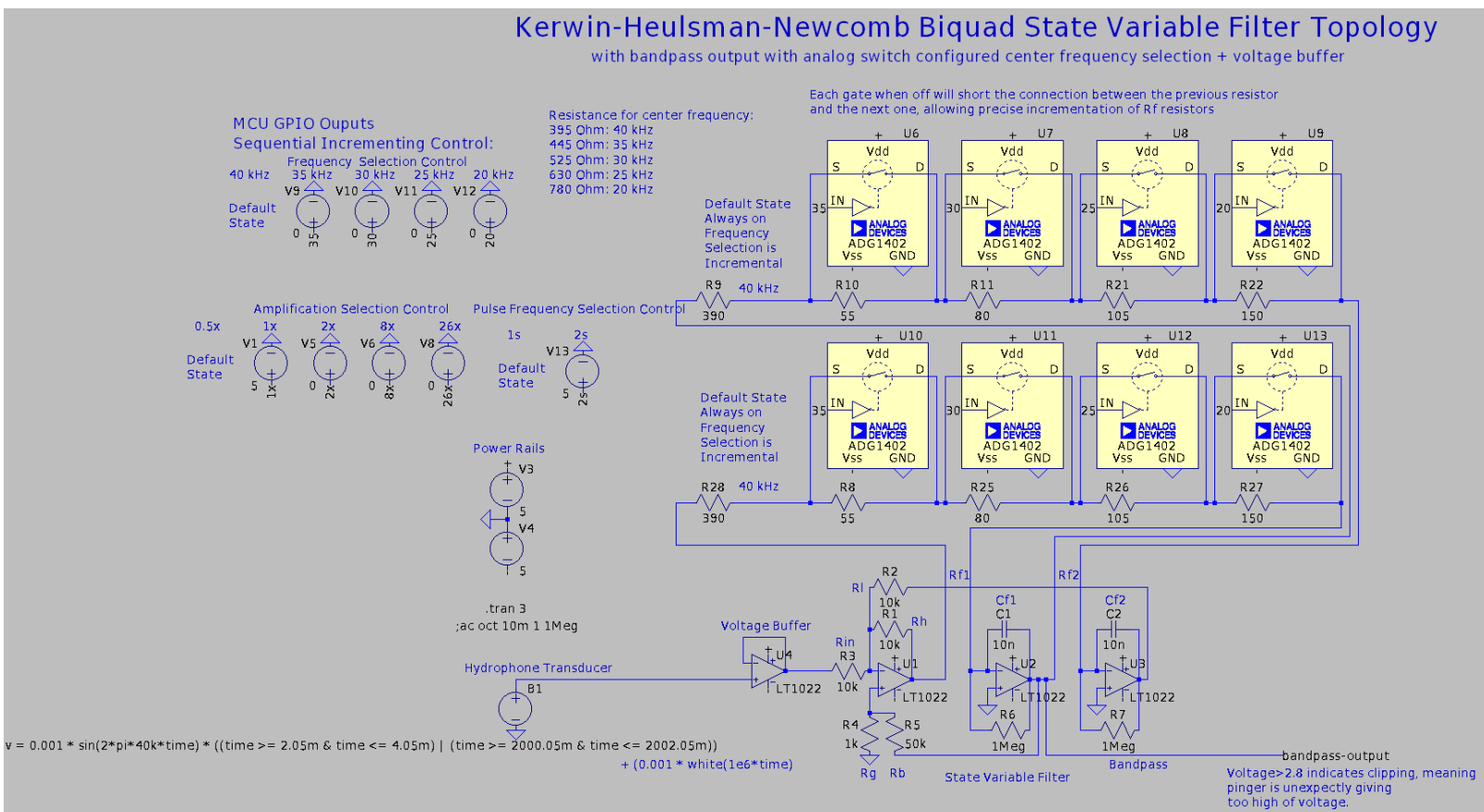
Dane Nicol

Designed by Dane Nicol

@Mechatronics SDSU

March 3, 2026

Block 1 (Voltage Buffer + SVF):

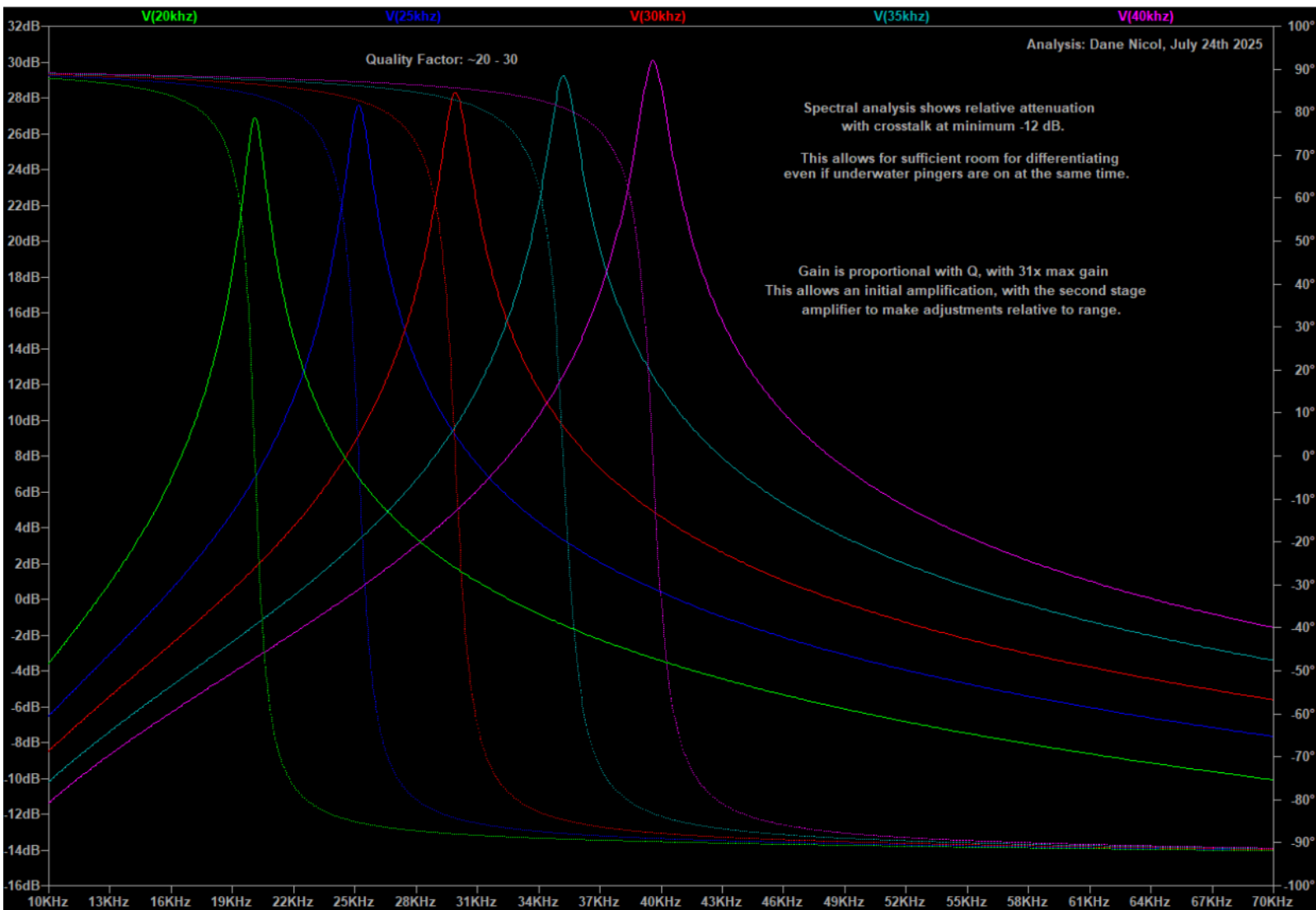


Input Buffer Stage:

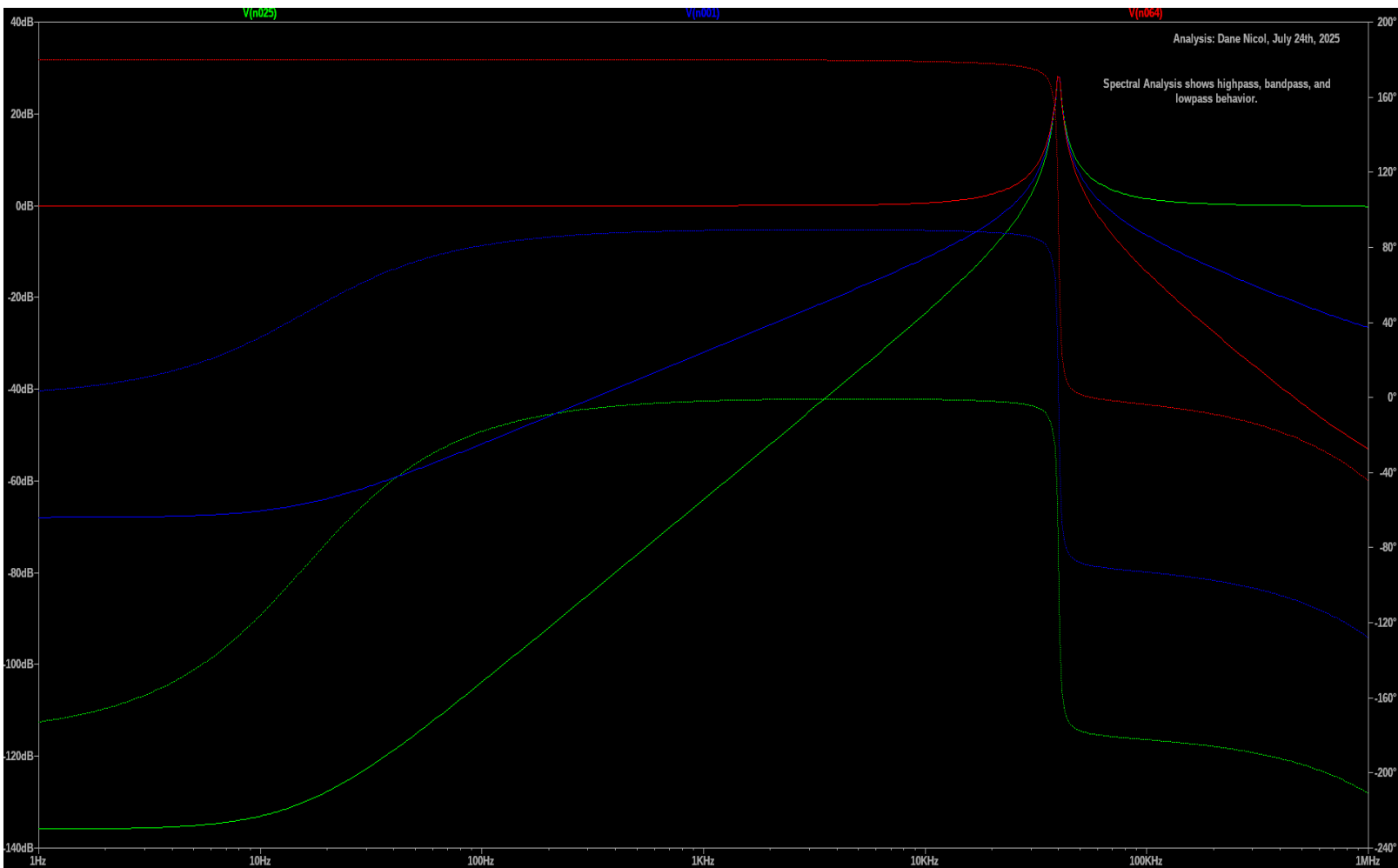
The hydrophone output is first routed into a voltage buffer. This stage isolates the transducer from downstream loading effects, ensuring the best signal integrity and preserving amplitude fidelity. This uses the LT1022, with ± 5 V rails. Bandpass output is already low-impedance, no additional buffer needed.

Analog Tunable State Variable Filter (SVF):

The buffered signal then feeds into the SVF, which is controlled via analog switches, that when off, short the series line of the resistors, bypassing what would be added resistance (that would decrease the selected frequency). The exact resistance was manually tested through many AC sweeps to determine that 395 Ω , 445 Ω , 525 Ω , 630 Ω , and 780 Ω all match the following frequencies 40 kHz, 35 kHz, 30 kHz, 25 kHz, and 20 kHz, respectively:



For the AC signal path, only bandpass behavior is required. However for full analysis of the state variable filter, the following AC sweep provides results for highpass, bandpass, and lowpass behavior:



Technical Analysis of SVF behavior:

This next section will focus on the transfer functions, and the derived behaviors of the transfer functions. All formulas will be specified and derived before complete analysis of the following frequencies: 20 kHz, 25 kHz, 30 kHz, 35 kHz, and 40 kHz.

The standard state variable filter implements the following three transfer functions in parallel:

$$\text{Low-pass (LPF): } H_{LP}(s) = \frac{\omega_0^2}{s^2 + (\omega_0/Q) * s + \omega_0^2}$$

$$\text{Band-pass (BPF): } H_{BP}(s) = \frac{(s * \omega_0)(1/Q)}{s^2 + (\omega_0/Q) * s + \omega_0^2}$$

$$\text{High-pass (HPF): } H_{HP}(s) = \frac{s^2}{s^2 + (\omega_0/Q) * s + \omega_0^2}$$

The bandpass transfer function is the focus for this analysis, since it is the only one going to be used.

Bandpass transfer function:

$$H_{BP}(s) = \frac{(s * \omega_0)(1/Q)}{s^2 + (\omega_0/Q) * s + \omega_0^2}$$

Parameters:

- $s \rightarrow$ Complex frequency domain variable: $s = \sigma + j\omega$
- $\omega_0 \rightarrow$ Natural (center) angular frequency
- $Q \rightarrow$ Quality factor (selectivity)

However, for our specified topology (Kerwin-Heulsman-Newcomb topology), the reciprocal of the quality factor is omitted. So the transfer function for the bandpass becomes:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + (\omega_0/Q) * s + \omega_0^2}$$

Before deriving each behavior, finding the angular frequency, quality factor, and bandwidth will be necessary:

Resonance Parameters:

Angular frequency:

$$\omega_0 = f_0 * 2\pi$$

- Both frequency and angular frequency are resonant

Quality Factor:

$$Q = \frac{\omega_0}{\Delta\omega}$$

- $\Delta\omega$: -3 dB bandwidth from resonant peak

Bandwidth:

$$BW = \frac{\omega_0}{Q}$$

Magnitude Response:

Substitute $s = j\omega$ (Note: $j^2 = -1$)

$$H_{BP}(j\omega) = \frac{\omega_0 j\omega}{j^2 \omega^2 + (\omega_0/Q) * j\omega + \omega_0^2} = \frac{\omega_0 j\omega}{-\omega^2 + (\omega_0/Q) * j\omega + \omega_0^2} = \frac{j\omega_0 \omega}{(\omega_0^2 - \omega^2) + j(\omega_0 \omega/Q)}$$

Let $\beta = \frac{\omega}{\omega_0}$ (normalized frequency):

$$H_{BP}(j\omega) = \frac{j\beta}{1-\beta^2+j(\beta/Q)}$$

Magnitude:

$$M(\omega) = |H(j\omega)| = \frac{|\beta|}{\sqrt{(1-\beta^2)^2 + (\frac{\beta}{Q})^2}}$$

Reverting to explicit angular frequency terms from β :

$$M(\omega) = \frac{\frac{\omega}{\omega_0}}{\sqrt{(1-\frac{\omega^2}{\omega_0^2})^2 + (\frac{\omega}{\omega_0 Q})^2}}$$

Example:

- Resonant frequency: $\omega_0 = 1000 \text{ rad/s}$ ($f_0 = 159 \text{ Hz}$)
- Input frequency: $\omega = 800 \text{ rad/s}$ ($f_0 = 127 \text{ Hz}$)
- Quality Factor: $Q = 5$
- $\beta = \frac{800}{1000} = 0.8$

Plugging values in:

$$M(\omega) = \frac{|0.8|}{\sqrt{(1-0.8^2)^2 + (\frac{0.8}{5})^2}} = 2.03x$$

Magnitude gain with initial values is 2.03, so at 159 Hz resonant frequency, 127 Hz is amplified by 2.03 times.

Gain Factor:

At resonance $s = j\omega_0$, the gain is:

$$|H_{BP}(j\omega_0)| = \left| \frac{\omega_0(j\omega_0)}{-\omega_0^2 + j(\omega_0^2/Q) + \omega_0^2} \right| = \left| \frac{j\omega_0^2}{j\omega_0^2/Q} \right| = Q$$

Peak gain $\propto Q$. This proportional correlation between gain and quality factor is not present in the standard textbook form, where it would instead be (peak gain = 1).

Poles:

The poles are the roots of the denominator:

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2 = 0$$

(has two poles, since degree = roots)

Using the quadratic formula:

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solving:

$$s = \frac{-(\omega_0/Q) \pm \sqrt{(\omega_0/Q)^2 - 4\omega_0^2}}{2} = \frac{-\omega_0}{2Q} \pm \omega_0 * \sqrt{\left(\frac{1}{2Q}\right)^2 - 1}$$

Let ζ = damping ratio:

$$\zeta = \frac{1}{2Q}$$

Then:

$$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$$

Underdamped: $Q > 0.5 \rightarrow \zeta < 1$

- Square root is imaginary $\rightarrow$ complex conjugate poles will result:

$$s = -\zeta\omega_0 \pm j\omega_0\sqrt{1 - \zeta^2}$$

- Impulse response looks like:

$$h(t) = e^{-\zeta\omega_0 t} * \sin((\omega_0\sqrt{1 - \zeta^2})(t))$$

- Slow decay with strong oscillatory response

Example: $\omega_0 = 1000$, $Q = 5$, $\zeta = \frac{1}{2Q} = 0.1$

$s = -100 \pm j994.987 \rightarrow$ rings at ~ 955 rad/s

Critically damped: $Q = 0.5 \rightarrow \zeta = 1$

- Square root is zero $\rightarrow$ repeated real pole:

$$s = -\omega_0 \text{ (double root)}$$

- Impulse response looks like:

$$h(t) = te^{-\omega_0 t}$$

- Fastest possible decay with non-oscillatory response

Example: $\omega_0 = 1000$, $Q = 0.5$, $\zeta = 1$

$s = -1000$ (twice)

Overdamped: $Q < 0.5 \rightarrow \zeta > 1$

- Square root is real $\rightarrow$ two distinct real poles:

$$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$$

- Both poles are going to be negative $\rightarrow$ sum of two decaying exponentials

- No oscillation, slower than critically damped

Example: $\omega_0 = 1000$, $Q = 0.2$, $\zeta = \frac{1}{2Q} = 2.5$

$$s = -2500 \pm j2291.288$$

$$s_1 = \sim -219$$

$$s_2 = \sim -4781$$

Stability:

Since all the poles have negative real parts for $Q > 0 \rightarrow$ system remains stable

However:

- If real part $< 0 \rightarrow$ system is stable
- If real part $= 0 \rightarrow$ system is marginally stable (sustained oscillations)
- If real part $> 0 \rightarrow$ system is unstable (output grows without bound)

This section for clarity will be under poles, BIBO stability section is specified under system properties.

Impulse response:

To find the impulse response specific to our topology, we can assume that the impulse response $h(t)$ is the inverse Laplace transform of the transfer:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

The impulse response can be found via:

$$h_{HP}(t) = \mathcal{L}^{-1}\{H_{BP}(s)\}$$

(Assuming a casual LTI)

The denominator: $s^2 + \frac{\omega_0}{Q}s + \omega_0^2$

Completing the square so it appears as: $(s + a)^2 + \beta^2$, since it has known inverse laplace transforms.

Grouping: $s^2 + \frac{\omega_0}{Q}s + \omega_0^2 = (s^2 + \frac{\omega_0}{Q}s) + \omega_0^2$

Completing the square on the quadratic in s: $s^2 + \frac{\omega_0}{Q}s = (s + \frac{\omega_0}{2Q})^2 - (\frac{\omega_0}{2Q})^2$

Therefore: $(s^2 + \frac{\omega_0}{Q}s) + \omega_0^2 = (s + \frac{\omega_0}{2Q})^2 - (\frac{\omega_0}{2Q})^2 + \omega_0^2$

Combining constants: $-(\frac{\omega_0}{2Q})^2 + \omega_0^2 = \omega_0^2(1 - \frac{1}{4Q^2})$

Defining:

- $a = \frac{\omega_0}{2Q}$

$$- \beta = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}$$

Denominator is now: $s^2 + \frac{\omega_0}{Q}s + \omega_0^2 = (s + a)^2 - \beta^2$

Transfer function is now: $H_{BP}(s) = \frac{\omega_0 s}{(s+a)^2 + \beta^2}$

Now the numerator needs to be reshaped to fit for the laplace pairs:

Damped cosine: $\mathcal{L}^{-1}\left\{\frac{s+a}{(s+a)^2 + \omega_d^2}\right\} = e^{-at} \cos(\beta t)$

Damped sine: $\mathcal{L}^{-1}\left\{\frac{\beta}{(s+a)^2 + \beta^2}\right\} = e^{-at} \sin(\beta t)$

Rewriting s in terms: $s = (s + a) - a$

$$H_{BP}(s) = \frac{\omega_0((s+a)-a)}{(s+a)^2 + \beta^2} \rightarrow \text{Distribute } \omega_0 \rightarrow \frac{\omega_0(s+a) - \omega_0 a}{(s+a)^2 + \beta^2}$$

Using the basic fraction property: $\frac{A-B}{D} = \frac{A}{D} - \frac{B}{D}$

Where:

$$- A = \omega_0(s + a)$$

$$- B = \omega_0 a$$

$$- D = (s + a)^2 + \beta^2$$

Now the transfer function becomes:

$$\frac{\omega_0(s+a)}{(s+a)^2 + \beta^2} - \frac{\omega_0 a}{(s+a)^2 + \beta^2}$$

Term 1 - Term 2

Term 1 corresponds to the damped cosine laplace pair:

$$\omega_0 * \mathcal{L}^{-1}\left\{\frac{(s+a)}{(s+a)^2 + \beta^2}\right\} = \omega_0 e^{-at} \cos(\beta t)$$

Term 2 relates to the damped sine laplace pair:

$$\frac{\omega_0 a}{(s+a)^2 + \beta^2} = \frac{\omega_0 a}{\beta} * \frac{\beta}{(s+a)^2 + \beta^2}$$

$$\frac{\omega_0 a}{\beta} * \mathcal{L}^{-1}\left\{\frac{\beta}{(s+a)^2 + \beta^2}\right\} = \frac{\omega_0 a}{\beta} e^{-at} \sin(\beta t)$$

Put terms back into their initial subtraction:

$$\omega_0 e^{-at} \cos(\beta t) - \frac{\omega_0 a}{\beta} e^{-at} \sin(\beta t)$$

Since the system is casual add u(t) and make exponential external:

$$h_{BP}(t) = e^{-at} [\omega_0 \cos(\beta t) - \frac{\omega_0 a}{\beta} \sin(\beta t)] u(t)$$

Our final impulse response:

$$h_{BP}(t) = e^{-at} \left[\omega_0 \cos(\beta t) - \frac{\omega_0 a}{\beta} \sin(\beta t) \right] u(t)$$

Solving an example: $\omega_0 = 1000$, $Q = 5$

First compute a and β :

$$\begin{aligned} - a &= \frac{\omega_0}{2Q} \rightarrow \frac{1000}{2 \cdot 5} = 100 \text{ s}^{-1} \\ - \beta &= \omega_0 \sqrt{1 - \frac{1}{4Q^2}} \rightarrow 1000 \sqrt{1 - \frac{1}{4(5)^2}} = 994.99 \text{ rad/s} \approx 995 \text{ rad/s} \end{aligned}$$

Plugging values in:

$$h_{BP}(t) = e^{-100t} [1000 \cos(995t) - 100.5 \sin(995t)] u(t)$$

At the given example values:

- Initial velocity: 1000, system is sensitive by sharp transients, instant, hard response, system also sensitive to high-frequency components.
- Envelope: e^{-100t} . Oscillations decay with time constant $1/100 = 0.01$ s. Rings for a short time. If Q increased, decay would be slower, vice versa.
- Oscillation frequency: ~ 995 rad/s. Frequency system rings at after impulse, slightly lower than resonant angular frequency ω_0 because the higher quality factor creates moderate damping.

Phase Response:

To find phase response, substitute the transfer function with $s = j\omega$. However we already did this for the magnitude response, hence:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Substituting ($s = j\omega$):

$$H_{BP}(s) = \frac{j\omega_0 \omega}{(\omega_0^2 - \omega^2) + j\frac{\omega_0}{Q}}$$

This complex number can be represented as a vector in the complex plane, via:

$$\phi(\omega) = \angle H_{BP}(j\omega) = \tan^{-1} \left(\frac{\text{Imaginary Part}}{\text{Real Part}} \right)$$

Since both numerator and denominator are both complex, total phase:

$$\phi(\omega) = \angle(\text{numerator}) - \angle(\text{denominator})$$

- Numerator is completely imaginary so $\rightarrow +90^\circ$
- Denominator has both real & imaginary $\rightarrow$ depends on angular frequency

Total phase:

$$\phi(\omega) = 90^\circ - \tan^{-1}\left(\frac{\frac{\omega_0}{Q}\omega}{\omega_0^2 - \omega^2}\right)$$

- At $\omega = \omega_0$ (resonant frequency), the phase shift is 0° , meaning the output signal is in phase with the input signal.
- At $\omega \rightarrow 0$ (below resonant frequency), the output leads the input by $+90^\circ$
- At $\omega \rightarrow \infty$ (above resonant frequency), the output lags the input by -90°

Solving an example: ($\omega_0 = 1000$, $\omega = 800$, $Q = 5$)

$$\phi(\omega) = 90^\circ - \tan^{-1}\left(\frac{\frac{1000}{5}800}{1000^2 - 800^2}\right) = 66.04^\circ$$

Output leads the input by 66 degrees.

Group Delay:

Solving for group delay is the negative derivative of the phase response with respect to frequency:

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}$$

Using the the derived phase response we got before:

$$\phi(\omega) = 90^\circ - \tan^{-1}\left(\frac{\frac{\omega_0}{Q}\omega}{\omega_0^2 - \omega^2}\right)$$

Group delay is derived by differentiating the above expression:

$$\frac{d}{d\omega} [\tan^{-1}(u)] = \frac{1}{1+u^2} * \frac{du}{d\omega}$$

Where $u = \frac{\frac{\omega_0}{Q}\omega}{\omega_0^2 - \omega^2}$, computing provides the final group delay:

$$\tau_g(\omega) = \frac{\frac{\omega_0}{Q}(\omega_0^2 + \omega^2)}{(\omega_0^2 - \omega^2)^2 + (\frac{\omega_0}{Q}\omega)^2}, \text{ showing peak delay when } \omega \neq \omega_0$$

At $\omega = \omega_0$, this expression simplifies to:

$$\tau_g(\omega) = \frac{Q}{\omega_0}, \text{ showing peak delay at resonance}$$

Solving an example: At $\omega = \omega_0$

With: $\omega_0 = 1000$, $Q = 5$

$\tau_g(\omega) = \frac{Q}{\omega_0} \rightarrow \frac{5}{1000} = 5 \text{ ms}$ (Envelope rises to peak with time constant of 5 ms from bust sine wave)

Solving an example: At $\omega \neq \omega_0$

With: $\omega_0 = 1000$, $\omega_0 = 800$, $Q = 5$

$\tau_g(\omega) = \frac{\frac{1000}{5}(1000^2 + 800^2)}{(1000^2 - 800^2)^2 + (\frac{1000}{5}800)^2} = \sim 12.8 \text{ ms}$ (Envelope rises to peak with time constant of ~ 12.8 ms from bust sine wave)

Asymptotic Roll-off:

The idealized rate at which a filter's gain decreases beyond its cutoff frequency:
From order under system properties, the system was determined to be second-order. This would automatically insinuate an asymptotic roll-off of -40 dB/decade or -12 dB/octave. However because of the bandpass coming from the first integrator output, roll-off shows behavior of a first-order system, giving:
-20 dB/decade or -12 dB/octave

Dynamic Model Analysis:

Ordinary differential equation (ODE) derivation:

Starting from the transfer function:

$$H_{BP}(s) = \frac{Y(s)}{X(s)} = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Cross multiplying:

$$Y(s)(s^2 + \frac{\omega_0}{Q}s + \omega_0^2) = X(s) * \omega_0 s$$

Applying inverse Laplace transform to each term:

$$\mathcal{L}^{-1}\{s^2 Y(s)\} \rightarrow \frac{d^2 y}{dt^2}$$

$$\mathcal{L}^{-1}\{s Y(s)\} \rightarrow \frac{dy}{dt}$$

$$\mathcal{L}^{-1}\{Y(s)\} \rightarrow y(t)$$

$$\mathcal{L}^{-1}\{s X(s)\} \rightarrow \frac{dx}{dt}$$

This provides the differential expression:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

This expression shows how the output $y(t)$ evolves over time in response to changes in the input $x(t)$.

Reverting back to transfer function:

Starting from the ODE:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

Applying Laplace transform to each term:

$$\mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} \rightarrow s^2 Y(s)$$

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} \rightarrow sY(s)$$

$$\mathcal{L}\{y(t)\} \rightarrow Y(s)$$

$$\mathcal{L}\left\{\frac{dx}{dt}\right\} \rightarrow sX(s)$$

This provides the s-domain expression:

$$Y(s)(s^2 + \frac{\omega_0}{Q}s + \omega_0^2) = X(s) * \omega_0 s$$

Factoring $Y(s)$ and solving for $H(s) = \frac{Y(s)}{X(s)}$:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Recovering the original transfer function.

System properties:

Causality:

From transfer function:

$$H_{BP}(s) = \frac{Y(s)}{X(s)} = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Checking if proper (degree of numerator $\leq$ degree of denominator)

- Numerator: $\omega_0 s \rightarrow \text{degree} = 1$
- Denominator: $s^2 + \frac{\omega_0}{Q}s + \omega_0^2 \rightarrow \text{degree} = 2$

This indicates a proper function, meaning it is causal. (Would be improper if numerator degree $>$ denominator, meaning it would be non-causal).

Checking poles (transfer function $\rightarrow \infty$) and zeros (transfer function = 0):

Zeros (What values of s make the numerator zero?):

- $s = 0 \rightarrow \omega_0 s = 0$

There is 1 zero.

Poles (What values of s make the denominator zero?):

- Poles = roots of denominator polynomial
- Polynomial of degree n has exactly n roots (real or complex)

Since we determined from the transfer function that the denominator has degree 2, it can be assumed that there are 2 poles.

This indicates a causality, because:

Number of finite zeros $\leq$ Number of finite poles

In our case 1 zero $\leq$ 2 poles.

From differential equation:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

This automatically shows a causal system because:

- Right-hand side (input): $\omega_0 \frac{dx}{dt} \rightarrow$ depends on the current rate of change of input.
- No future values of $x(t)$ are needed, only present and past values.

Memory:

- A system has memory if its output depends on past (or future) inputs.
- A system is memoryless if output depends only on the current input.

From transfer function:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

- If the transfer function does not have poles or zeros (just a constant) then it is memoryless.
- If the transfer function has any poles or zeros, it has memory.

We just checked previously if the system had zeros or poles (which it does), so it can be assumed the system has memory from its transfer function.

From differential equation:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

- Left side contains derivatives of $y(t) \rightarrow$ system state depends on past behavior.
- Right side has $dx/dt \rightarrow$ response depends on how input is changing, not just current value

Automatically can be inferred from the ODE that the system has memory.

BIBO stability:

The system is BIBO (bounded-input bounded-output) stable if every bounded input produces a bounded output.

For LTI systems, BIBO stability holds only if the impulse response is absolutely integrable:

- Must decay fast enough over time (has exponential decay such as e^{-at}).
- Cannot grow with time/does not decay (constant or oscillatory without decay).
- Slow decay still can diverge making it non-integratable.

For our impulse response:

$$h_{BP}(t) = e^{-at} \left[\omega_0 \cos(\beta t) - \frac{\omega_0 a}{\beta} \sin(\beta t) \right] u(t)$$

Exponential decay shows that it is integratable, hence the system is BIBO stable.

In the s-domain: all poles must lie in the open left half-plane (negative real parts).
Already solved in stability under poles.

Linearity:

This can already be assumed based on the transfer function being an LTI, however from the ODE we can figure out if it satisfies superposition:

- Additivity: $y(x_1 + x_2) = y(x_1) + y(x_2)$
- Homogeneity: $y * (cx) = cy(x)$
- Combined, they form superposition: $y(c_1 x_1 + c_2 x_2) = c_1 y(x_1) + c_2 y(x_2)$

Our ODE:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0^2 * \frac{dx}{dt}$$

- All terms involving $y(t)$ and $x(t)$, and their derivatives are linear (no y^2 , $y \frac{dy}{dt}$).
- Zero initial conditions make it linear (non-zero initial conditions would make it non-linear).
- The differential operator on the left is linear $\rightarrow$ satisfies both additivity and homogeneity.

The system is linear.

Order:

The order of a continuous-time LTI system is the order of the differential equation describing it, or equivalently, the degree of the denominator polynomial in the transfer function.

From transfer function:

$$H_{BP}(s) = \frac{\omega_0 s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

Degree of denominator = 2

Therefore it is a second-order system.

From differential equation:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

Second derivative = 2

Therefore it is a second-order system.

Note: because of the topology of state variable filters, the bandpass output is taken from the first integrator stage, however a single integrator acts as a first order circuit on its own. Despite this, the BP signal is still part of a second-order feedback loop (it's fed back to influence the input, creating the full second-order dynamics).

Time Invariance:

From differential equation:

$$\frac{d^2 y}{dt^2} + \frac{\omega_0}{Q} * \frac{dy}{dt} + \omega_0^2 y = \omega_0 * \frac{dx}{dt}$$

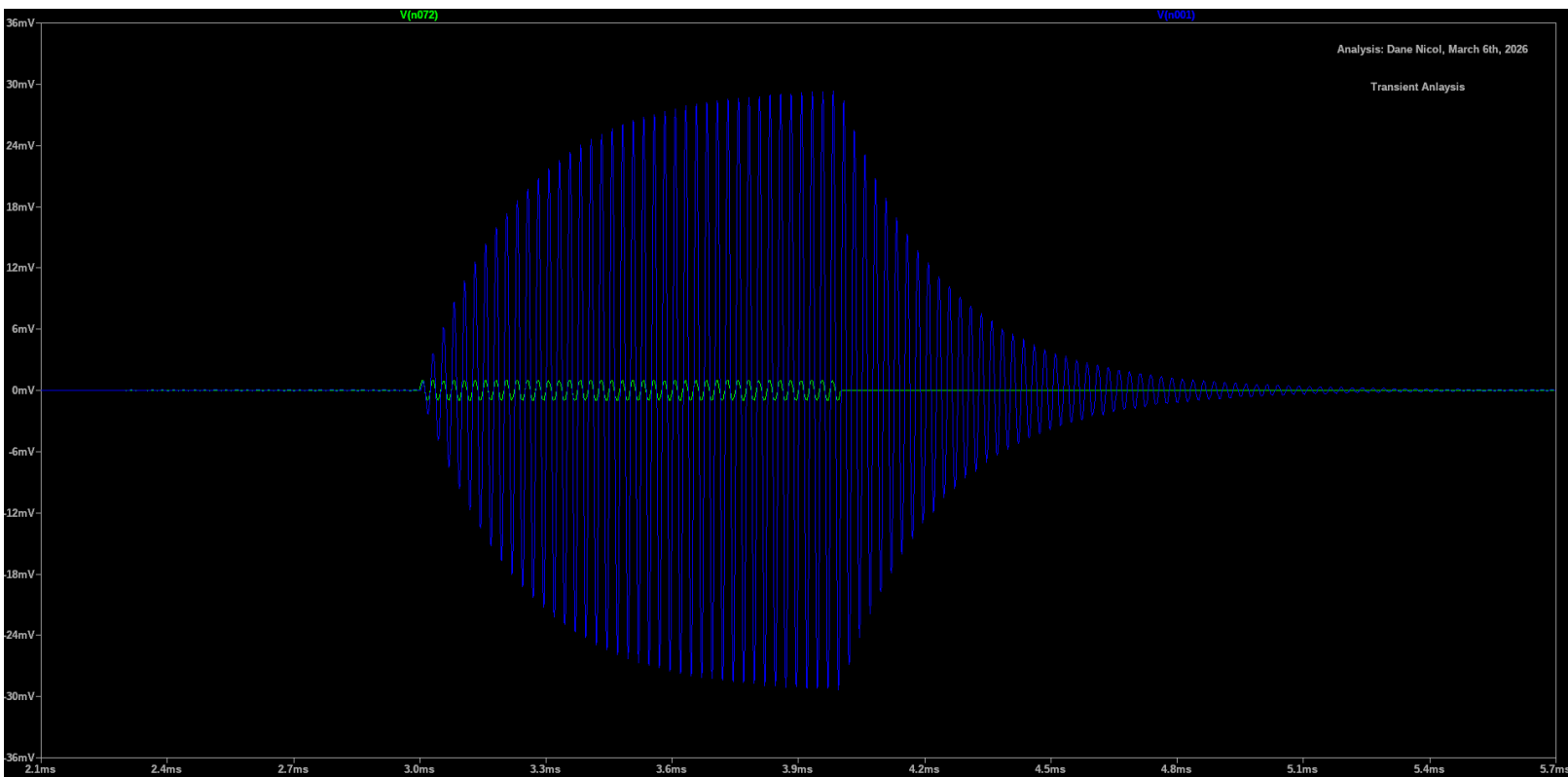
The system only has constant coefficients (no terms involving t , t^2 , or e^t). Note: derivative terms do not affect the time invariance, coefficients of the ODE are still not functions of time. Therefore it is time invariant.

Simulation Full Analysis:

(at 40 kHz resonance)

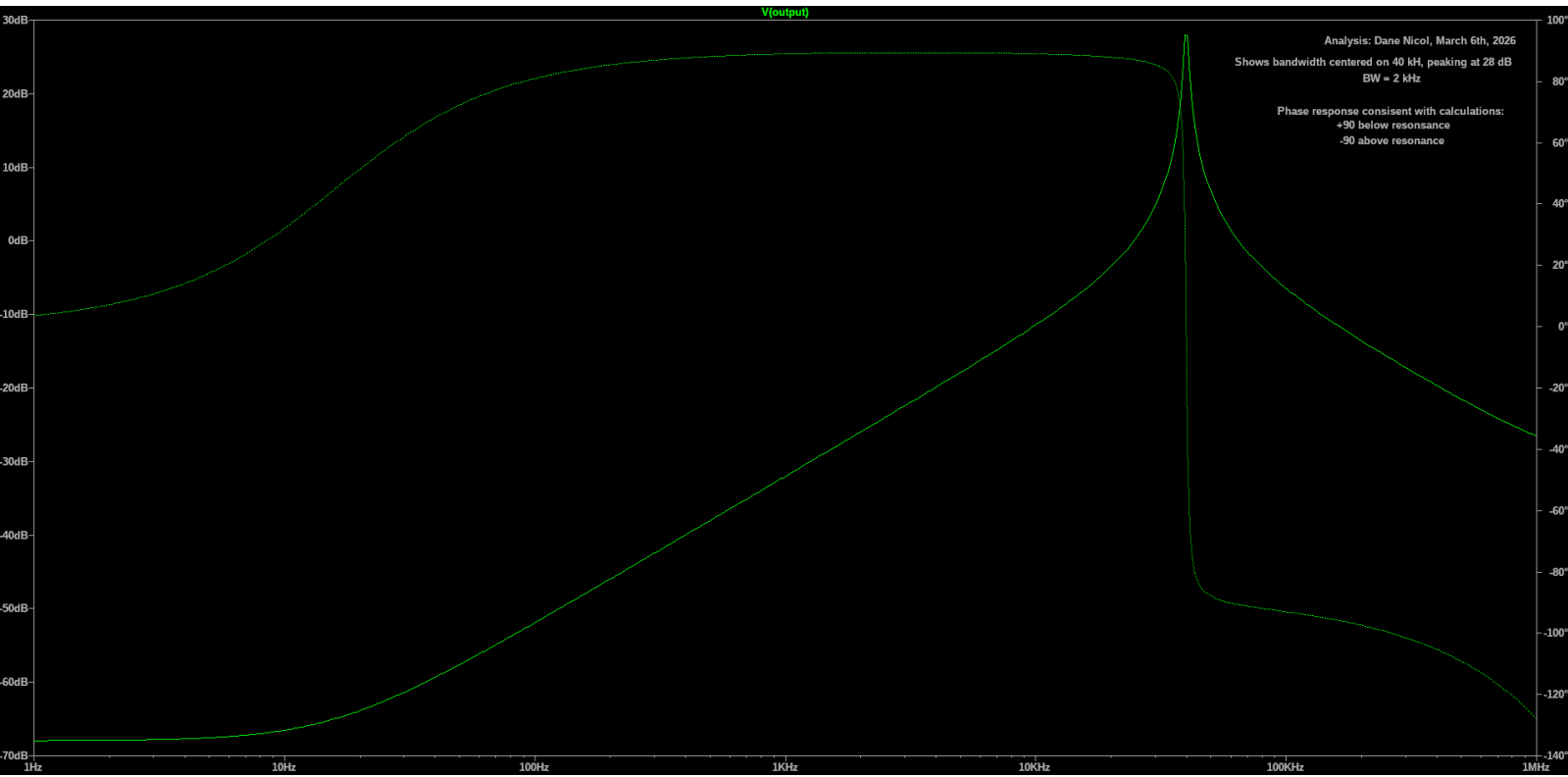
Transient Analysis:

Group delay shows appropriate delay, damping shows underdamped behavior, gain factor is proportional to Q. All calculations consistent with transient analysis.

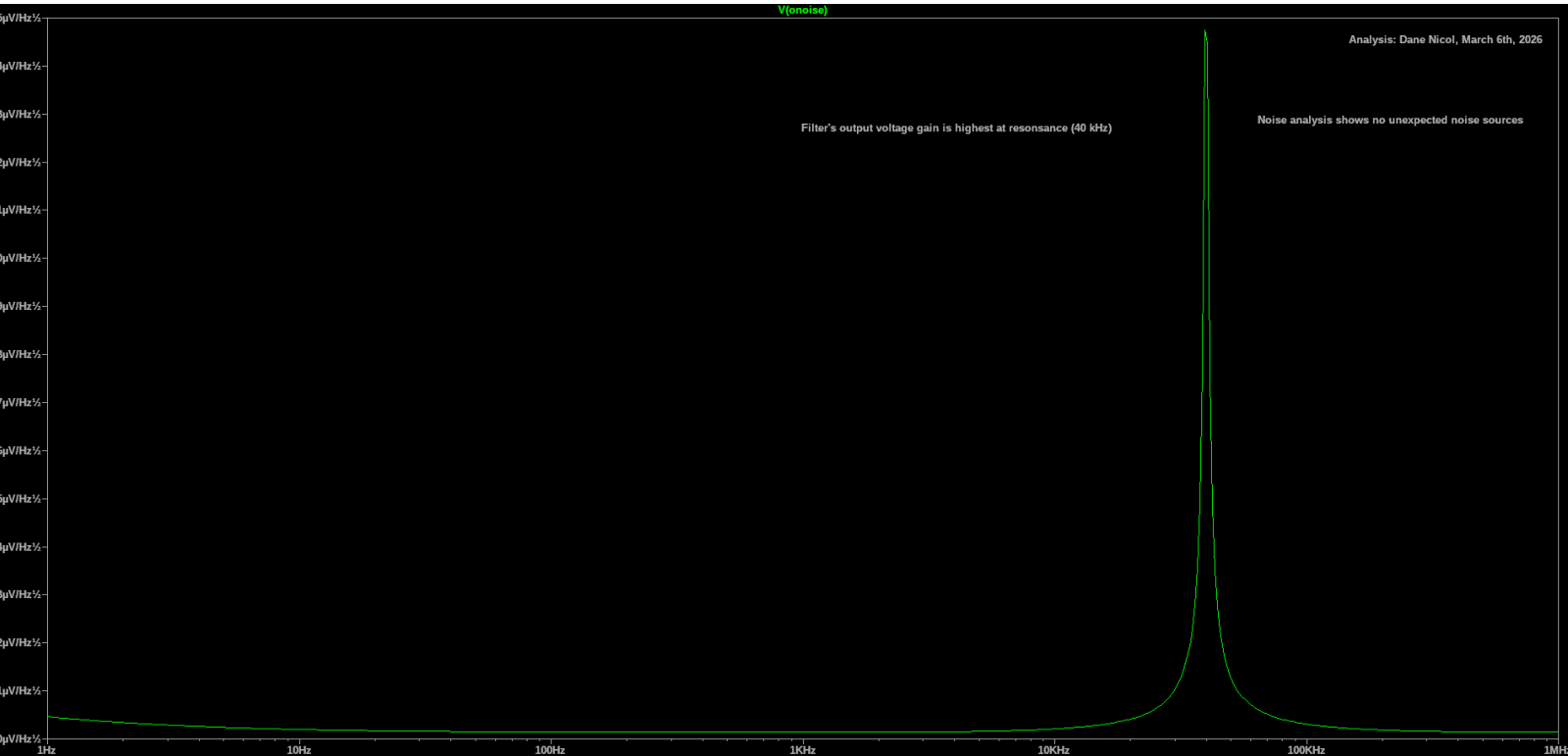


Spectral Analysis:

Bandwidth, phase response, quality factor, and peak all consistent with previous calculations.

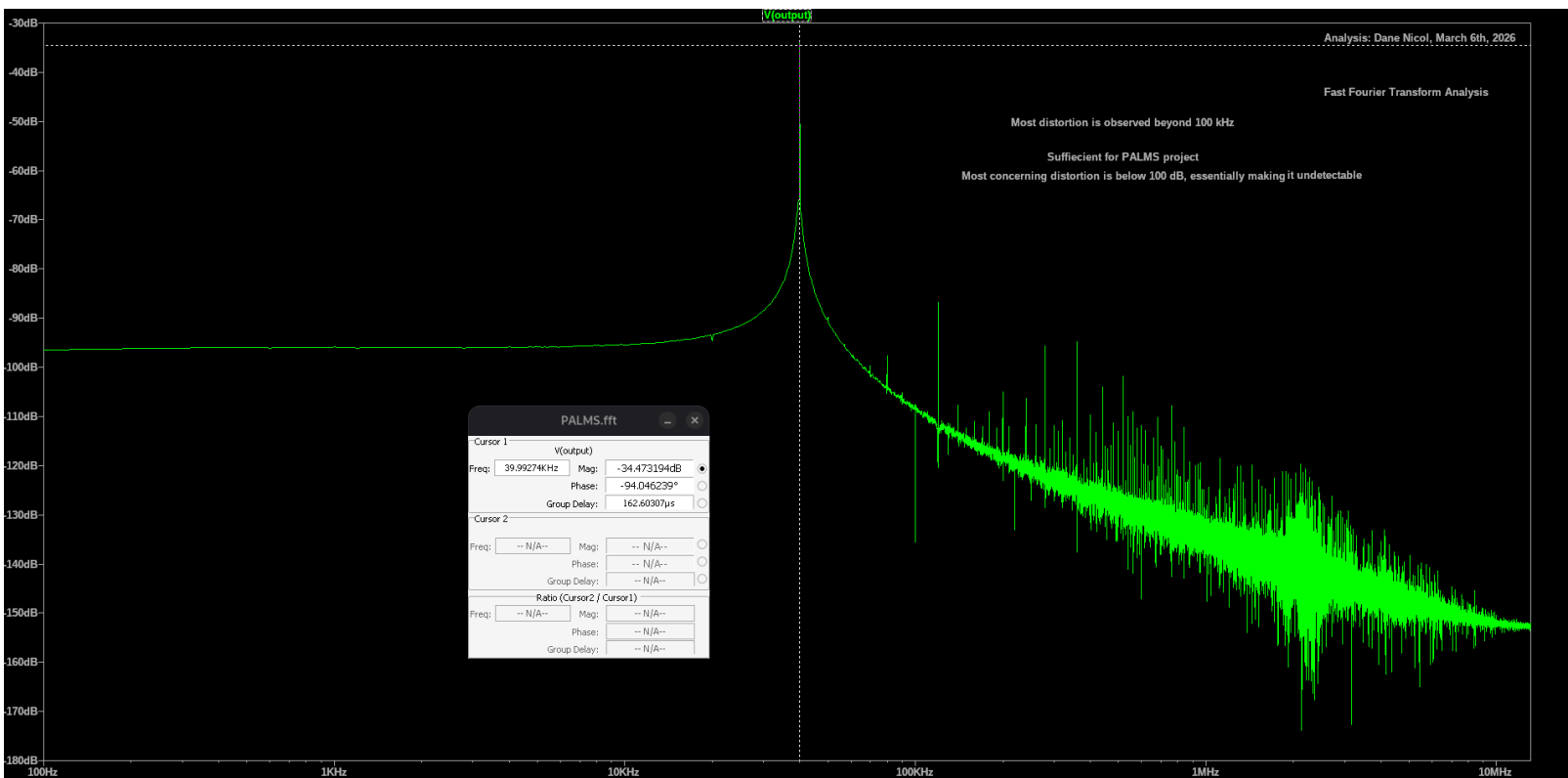


Noise Analysis:
Showing no unexpected noise sources. Highest voltage gain centered at resonance.



Fourier Analysis:

Fast fourier transform graph (FFT):



Error log (for harmonic data):

N-Period=1

Fourier components of V(output)

DC component:2.49711e-006

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]	Normalized Phase [deg]
1	4.000e+04	2.941e-02	1.000e+00	-16.96°	0.00°
2	8.000e+04	1.468e-05	4.993e-04	-163.84°	-146.88°
3	1.200e+05	4.899e-05	1.666e-03	43.00°	59.96°
4	1.600e+05	1.237e-05	4.206e-04	-44.25°	-27.29°
5	2.000e+05	7.653e-06	2.602e-04	-147.10°	-130.13°
6	2.400e+05	8.780e-06	2.986e-04	-92.69°	-75.73°
7	2.800e+05	1.771e-05	6.023e-04	147.07°	164.03°
8	3.200e+05	4.008e-06	1.363e-04	139.53°	156.50°
9	3.600e+05	2.202e-05	7.486e-04	70.35°	87.31°

Total Harmonic Distortion: 0.207372%(0.223797%)

Date: Fri Mar 06 14:48:46 2026

Total elapsed time: 27.635 seconds.

Interpretation:

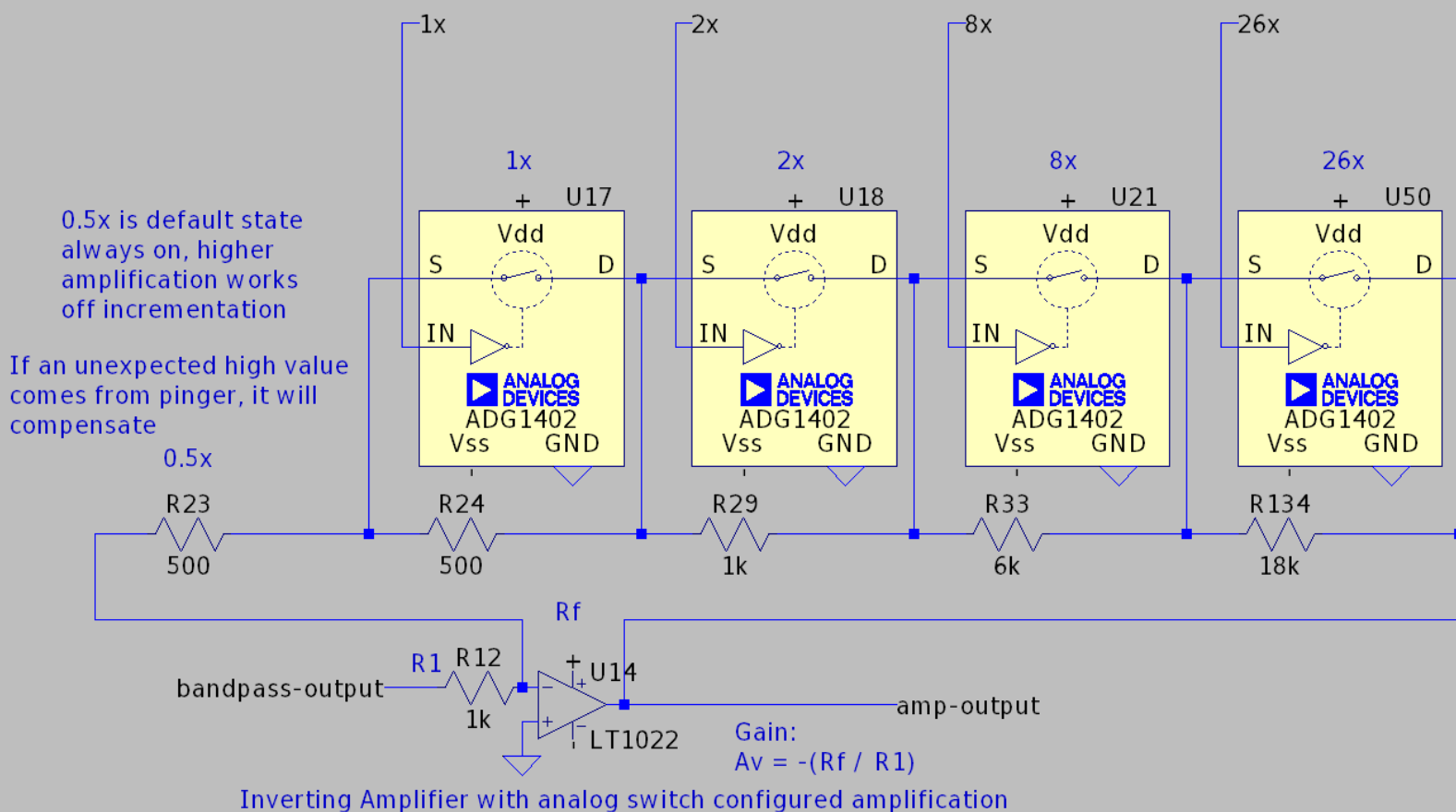
Very low and indicates excellent signal purity.

Below 0.1%: Generally inaudible, considered high fidelity.

0.1% – 1%: Good to acceptable; 0.2% is well within this range and typically not perceptible in most applications.

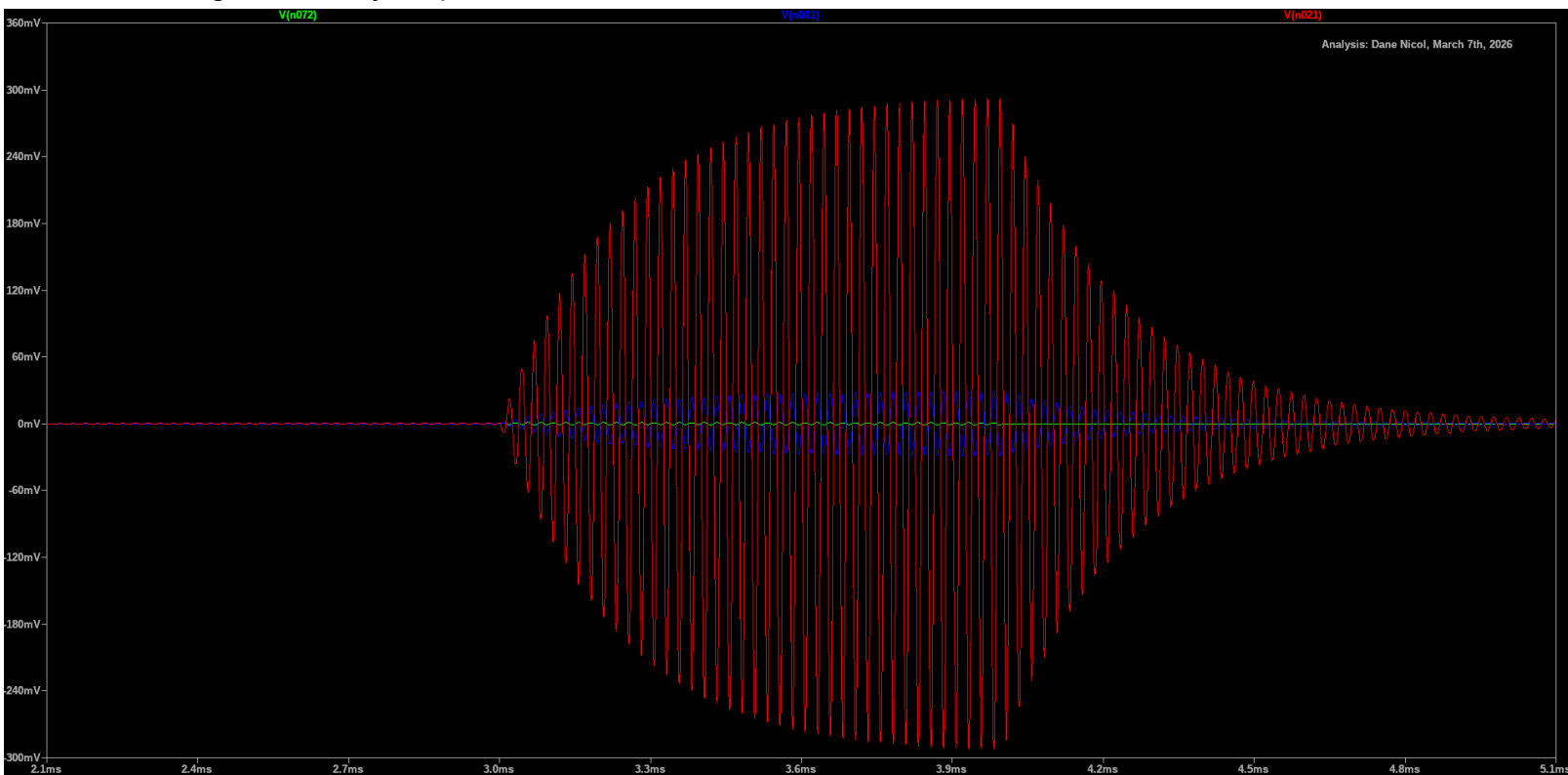
Block 2 (Manual Amplifier):

Manual Amplification Adjustment



The following circuit is the manual amplifier, a simple inverting amplifier with the same analog switch configuration the SVF had. The ideal state should leave amplification at 1x, unless the situation requires (2x, 8x, 26x) (weaker signal/longer range), or 0.5x (overamplification via SVF, requires reduction to prevent wave clipping if signal is stronger than assumed).

The following analysis shows an example of a 1 mV, 40 kHz waveform (while SVF has its center frequency at 40 kHz), being amplified by the SVF to 30 mV, then proceeding to get manually amplified 10x to 300 mV:



This amplification stage is less of a necessity and more of a precaution, as pulse detection has a dynamic threshold.

Voltage gain of the inverting amplifier configuration:

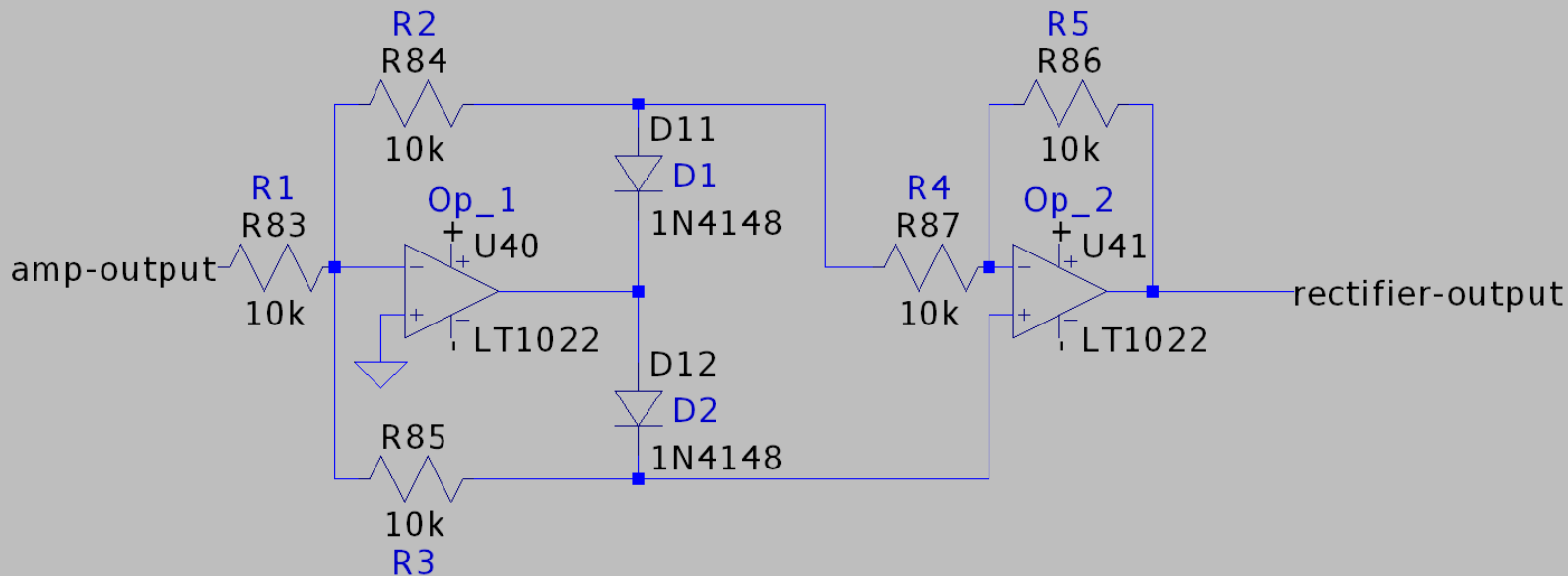
$$G = - \frac{R_f}{R_1}$$

Worth noting, the output signal of the inverting configuration is 180° out of phase relative to the input signal. This should not affect TDOA.

Block 3 (Rectifier Stage):

Precision Full Wave Rectifier

The Precision Full Wave Rectifier passes through positive waves with unity gain and inverts negative waves with unity gain, essentially providing a rectified response using diode biasing as control.



The next stage is a precision full wave rectifier, for the purpose of removing negative voltages in preparation for amplitude-threshold/delay processing.

This rectifier:

- When $V_{in} > 0$, op amp output goes negative, forward biasing diode 1 and reverse biasing diode 2. This creates an equivalency of two proportional inverting amplifiers connected together in series:

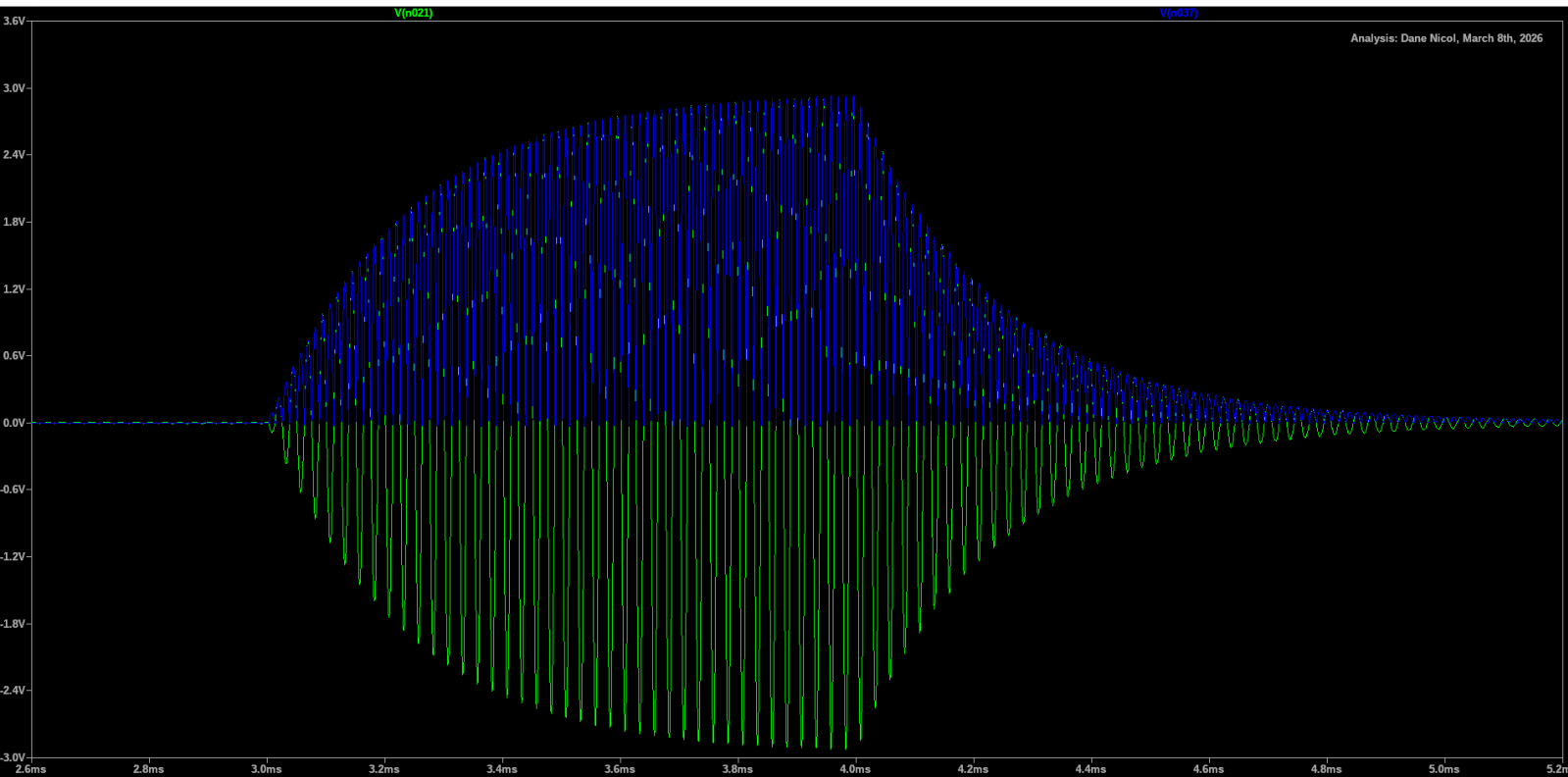
$$V_{out} = \left(-\frac{R2}{R1}\right) * \left(-\frac{R5}{R4}\right) = V_{in} \text{ (assuming 10k constant for resistors).}$$

This allows the positive wave signals to pass through what is essentially an non-inverting amplifier with 1x gain (unity voltage buffer).

- When $V_{in} < 0$, op amp output goes positive, reverse biasing diode 1 and forward biasing diode 2. This creates an equivalency of a singular inverting amplifier with 1x gain, flipping the negative waves positive:

$$V_{out} = \left(-\frac{R5}{R4}\right)$$

The following analysis shows a 40 kHz, 300 mV sine wave be rectified:



As shown, negative wave cycles are flipped to positive.

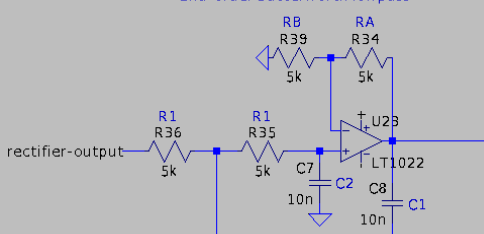
Block 4 (Amplitude-Threshold/Decay):

Amplitude-Threshold/ Delay Block

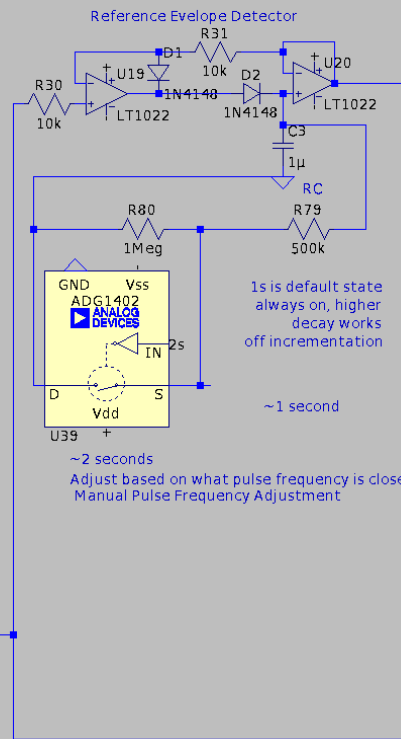
Envelope Detector holds DC value of peak input, with a decay rate of:
 $V(t) = V_0 * e^{(-t / RC)}$
 Where:
 V_0 = initial voltage
 t = time of discharge
 RC = resistor capacitor config

Note: discrepancy between formula and simulation is expected. Peak of envelope will also be either higher or lower than input.

2nd-order butterworth lowpass



The active lowpass filter does the initial smoothing for two inputs into a comparator:
 1: Reference decaying envelope
 2: Input signal

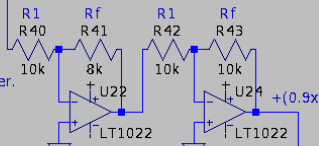


1s is default state always on, higher decay works off incrementation

~1 second

~2 seconds
 Adjust based on what pulse frequency is closer.
 Manual Pulse Frequency Adjustment

Cascaded inverting amplifiers:
 $-0.8x * -1x$ (non-inverting output)



Trigger Comparator A

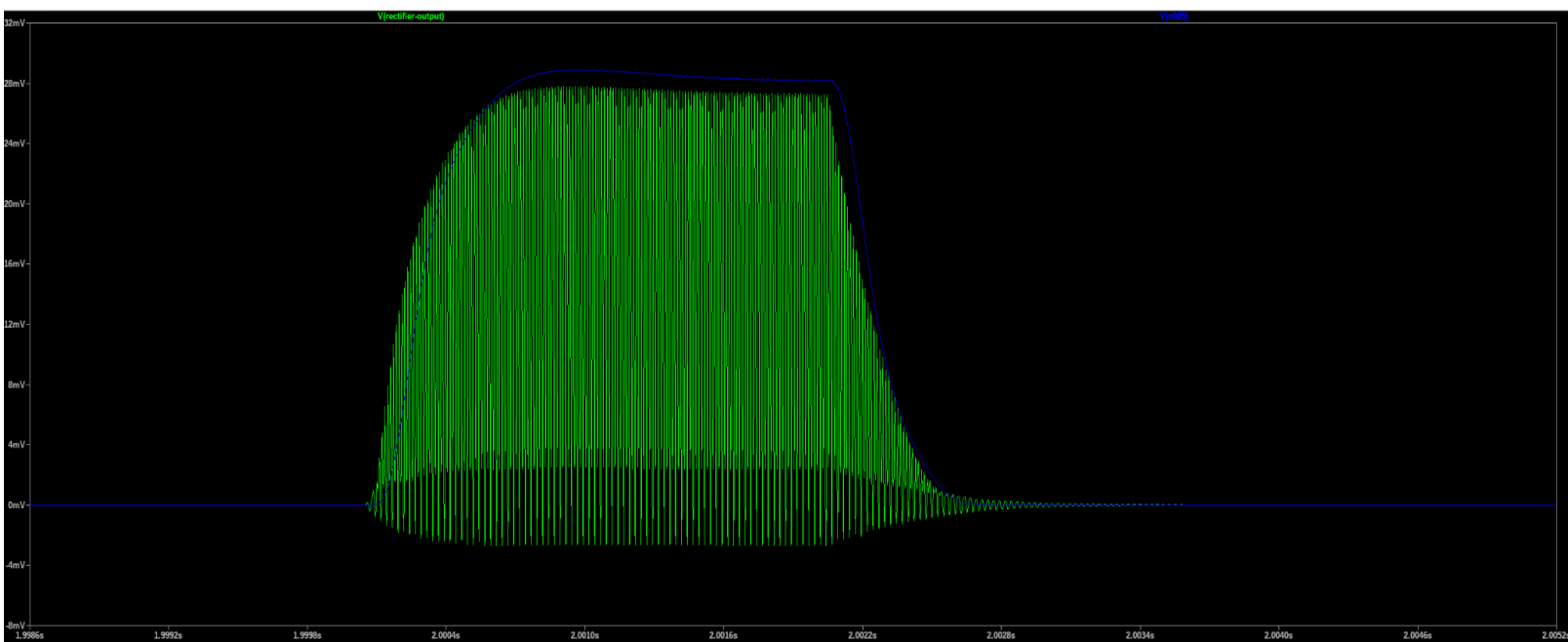
$V_{ref} \rightarrow$
 $V_{in} \rightarrow$
 comp-output

Triggers if >80% of the maximum amplitude of the reference is met with the input pulse from V_{in} .

This is the pulse that is inputted into the comparator, which is then compared against Envelope detector that holds the peak signal 80%, so if this signal is over that, the comparator activates.

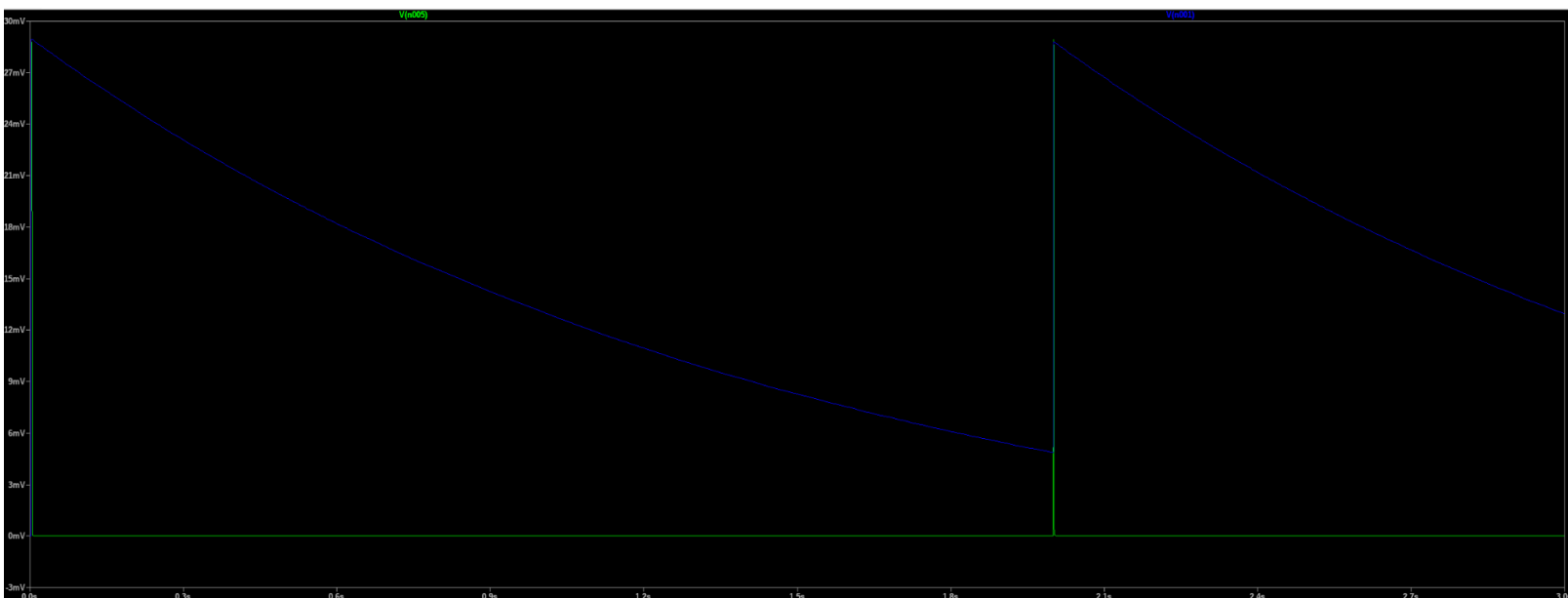
The same methods that I used before to analyze the transfer function of the bandpass and derive several behaviors from are omitted for sake of redundancy and brevity of this documentation.

Block 4 starts off with a 2nd-order butterworth lowpass filter to properly smooth the high frequency input signal into a pulse, as shown in the following transient:



Without it, it would rapidly oscillate the trigger comparator's input. As shown in the simulation, there is a slight, negligible delay that does not affect TDOA, as the other input for horizontal or vertical comparison also will have this delay. This should also be considered for all propagation delay present in the circuitry.

Following this smoothed pulse are the two inputs of the trigger comparator, one reference, one input signal. The input signal has the pulse go directly to it. For the reference, the pulse first goes through an envelope detector that holds the peak of the pulse, with a selectable decay (2 or 1 second decay). The following transient shows the behavior of the 2 second:



This provides a lot enough time for the largest magnitude signal of the selected pinger frequency (which should theoretically be the true signal of the pinger, nothing else), and holds it for enough time to be used as a voltage reference for the trigger comparator input signal, while decaying to allow for automatic threshold adjustment in regards to the magnitude.

The following formula solves for the approximate decay rate:

$$V(t) = V_0 * e^{-t/RC}$$

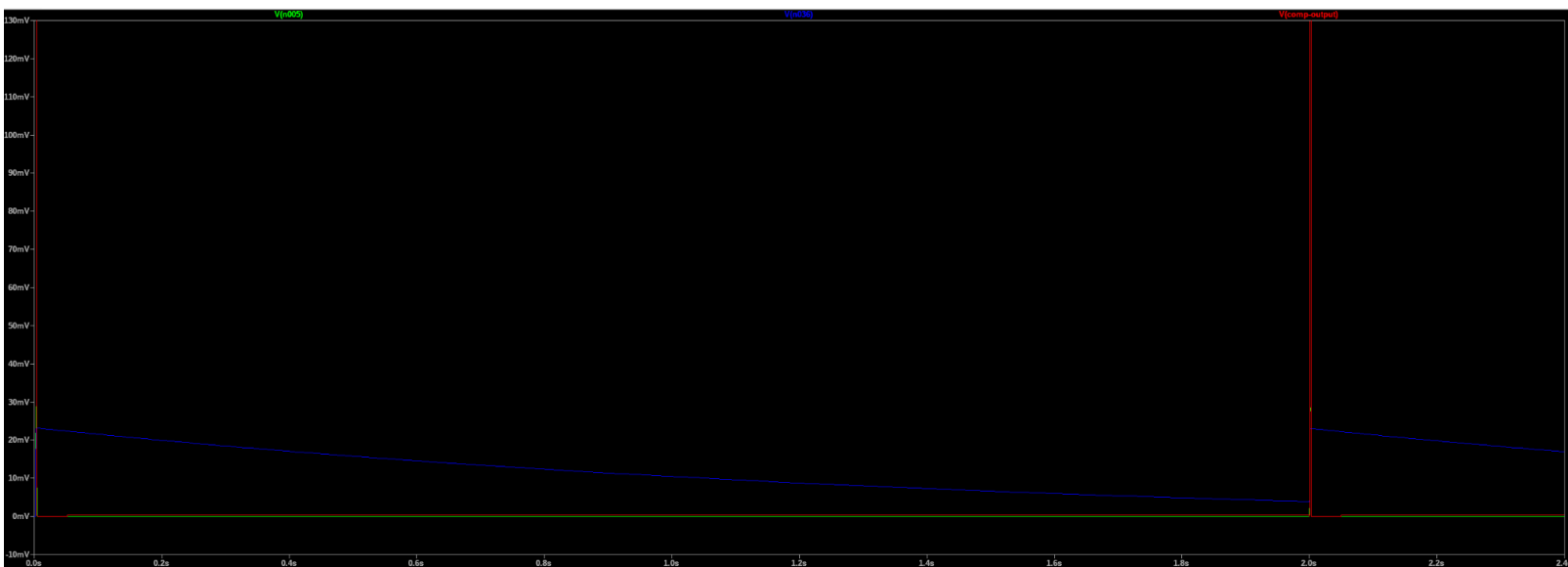
Where:

- V_0 is the initial voltage
- t is the time of discharge (so the very ns after the pulse)
- RC the resistor capacitor config

There will however be some non-unsignificant discrepancy between the formula and actual result, so following simulation for exact numbers is necessary.

Before this goes into the trigger comparator reference, there are cascaded inverting amplifiers that I have manually cycled through many variations to find what reduction in amplitude would be ideal to provide sufficient headroom for the input signal to prevent oscillation, since the signal will be “wobbly” under non-ideal conditions.

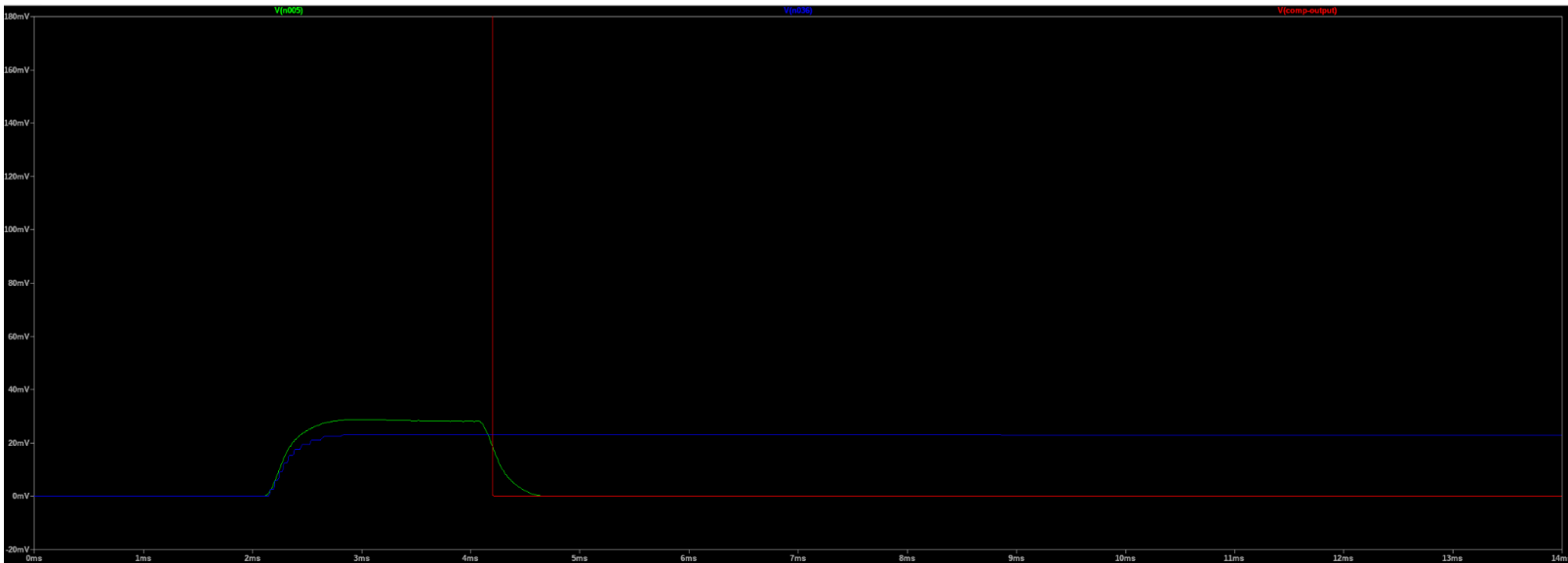
Now, the following transient shows the ~2.4 second duration of the behavior of the trigger comparator with 2 pulses from the pinger:



With no oscillation, the digital output is nearly 1:1 of the pulse of the pinger, finally providing a clean and accurate digital signal ready for TDOA calculations.

For clarification The red signal goes to 5 volts since it is a digital output, but this simulation graph was shortened to show the behavior of the other signals as well.

Another up close transient shows the behavior working perfectly for a clean digital response:



This is now ready to be input into a microcontroller or computer to calculate the exact angle of a pinger relative to the UAV, for both horizontal and vertical orientations, allowing localization regardless of the depth of the pinger in the water.

Block 5 (Arming/Disarming Control System):

Arming/Disarming Analog System Unit AB

Digital Control:

Using the Q output from the analog SR latch system, a high will indicate hydrophone A had received the last ping sooner than hydrophone B which means the AUV would need to turn into the direction of hydrophone A. Same logic for a low from Q output, indicating hydrophone B received a ping sooner than hydrophone A

Arming Comparator A

Precision Envelope Detector

Inverting Config, Disarming time changable with Vref & RC of envelope detector
Current config allows for ping pulse frequency > 50 ms.
Line will disarm other comparator under 50 ms.

Summer & Comparator (AND Logic) Unit A

Noninverting config:
 $2 \times 2 = 4$ (Vref)
(3.3V for Vref for headroom;
resistor dividers do not give perfect 2V output).

Analog SR latch system

Biases the comparator output from 5V to 2V in order for the AND logic to be functional

Noninverting config:
 $2 \times 2 = 4$ (Vref)
(3.3V for Vref for headroom;
resistor dividers do not give perfect 2V output).

Summer & Comparator (AND Logic) Unit B

Non-Inverting comparator

Current config allows for ping pulse frequency > 50 ms.
Line will disarm other comparator under 50 ms.

Precision Envelope Detector

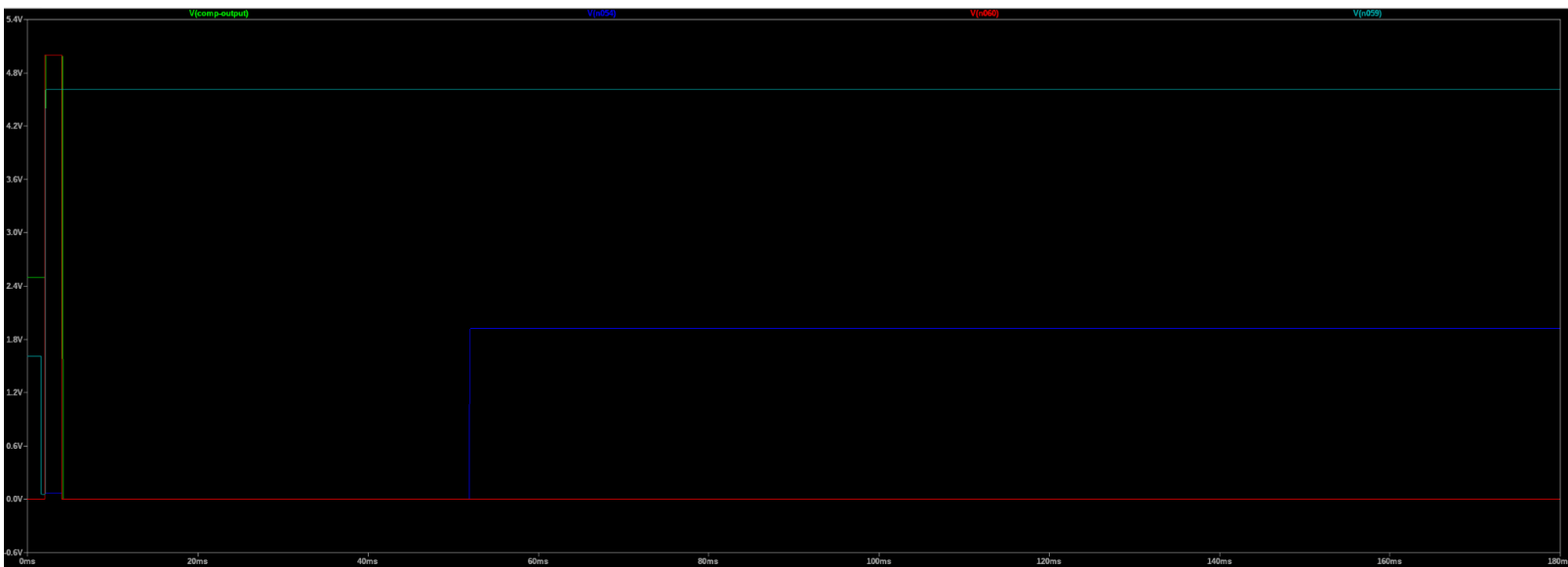
Arming Comparator B

Block 5 is an optional, but useful last bit of circuitry that allows the PALMS system to locate a pinger using **completely** analog circuitry! To describe the function:

For example, let's say that unit A had received a digital pulse before unit B. Unit A would have the pulse travel through the "AND logic" which already had its other input high since the inverted comparator on unit B's side would be still high, and would then proceed to activate the SR latch system, which would stay high unless reset by a pulse from unit B.

However, unit B would have received a pulse very shortly after and would reset the SR latch system. Because of the arming/disarming system, arming comparator A prevents that. The pulse unit A received would move through an envelope detector and hold that signal high for around 50 ms, going into the inverted arming comparator A, which would in turn go low, and disarm unit B's "AND logic". 50 ms is perfectly acceptable because the shortest pulse frequency of the pingers is 500 ms, or half a second. The same exact behavior would happen on the other side if unit B had received a pulse before unit A.

While much easier to understand the whole process by manually simulating transients, the following simulation shows multiple signals, with primary importance of the AND logic of unit B being turned off, and the SR latch holding a high, indicating unit A received a response before unit B, with 40 microseconds being the difference of time between unit A and B:



The circuitry shown throughout this documentation is easier to simply copy/paste, 3 analog processing modules (for 3 hydrophone transducers to compare both horizontal and vertical angle) and 2 TDOA modules (to output high/low to go left/right, and another high/low to go up/down).

This concludes the PALMS System Design documentation. This document is subject to change.